

3.2.3 How life has developed on Earth

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 8			
Darwin's theory of evolution states that all species of living things have evolved from simple life forms that first developed more than three billion years ago.	Evidence to support this theory of evolution is limited to fossil evidence and the similarity of characteristics between species.	4.6.2.2	4.4.4.2
In natural selection, individuals with characteristics most suited to their environment are most likely to survive to breed successfully.	An example is the distribution of the peppered moth.	4.6.2.2	4.4.4.2
Artificial selection (selective breeding) is the process by which humans breed plants and animals for particular genetic traits.	Examples include disease resistance in food crops, domestic dogs with a gentle nature and plants with large flowers.	4.6.2.3	4.4.4.5
Outcome 9			
There are two types of reproduction: 1. sexual reproduction, which involves the joining of male and female sex cells. There is a mixing of genetic information, which leads to variety in the offspring. 2. asexual reproduction, where only one individual is needed as a parent. There is no mixing of genetic information, which leads to identical offspring (clones).	The term 'gamete' is not required. Suggested activity for TDA Asexual reproduction: investigate how alike the plants grown from runners are, eg mint or strawberries.	4.6.1.1	4.4.3.1
Outcome 10			

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The genetic material in the nucleus of a cell is made of a chemical called DNA, which is contained in structures called chromosomes. Students should know that a cell consists of a nucleus that controls the actions of the cell, and cytoplasm.		4.6.1.3	4.4.3.1
Chromosomes carry genes that control the characteristics of the body. Humans have 23 pairs of chromosomes. Only one pair carries the genes that determine sex: females have the same sex chromosomes (XX); in males the chromosomes are different (XY).	Suggested activity for TDA Investigate whether or not two characteristics are linked, eg finger length and height. Diagrams of genetic crosses will not be required.	4.6.1.6	4.4.3.1
In genetic engineering, genes from chromosomes of humans and other organisms can be 'cut out' and transferred to the cells of other organisms. Students should be aware of the potential benefits and risks of genetic engineering.		4.6.2.4	4.4.4.6